Aryan Goel

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MS Data Science candidate turning messy business data into clear, decision-ready insights — SQL, Power BI, and real statistical rigor.

EDUCATION

University of Southern California — Los Angeles, CA
Master of Science, Computer Science (Data Science Specialization) — *Aug. 2025 – May 2027 (Expected)*
***Relevant Coursework**: Machine Learning for Data Science, Applied Statistical Inference, Deep Learning, Algorithms, Database Systems (indexing, storage engines, query optimization), Foundations of Data Management (Hadoop, Spark, distributed SQL), NLP*

Manipal University Jaipur — Jaipur, India
Bachelor of Technology, Computer & Communication Engineering — GPA: 9.15/10 — *July 2021 – May 2025*

TECHNICAL SKILLS

Languages: Python, SQL, R (basic)
Data Analysis & Visualization: Excel (pivot tables, VLOOKUP/INDEX-MATCH), Power BI (DAX, Power Query, data modeling), Tableau (familiarity), Matplotlib, Seaborn, Dashboarding & Reporting
Statistics & Experimentation: Hypothesis Testing, A/B Testing, Confidence Intervals, Descriptive Statistics, Confounding/Simpson's Paradox Analysis, Causal Reasoning
SQL & Databases: PostgreSQL, MySQL, DuckDB (OLAP/columnar), Window Functions, Query Optimization, EXPLAIN/EXPLAIN ANALYZE, ClickHouse, PySpark
Business Intelligence: Snowflake, KPI Dashboards, Stakeholder Reporting, ETL/ELT Fundamentals
Tools: Git, Jupyter, VS Code, Streamlit, Docker, scikit-learn, SHAP

EXPERIENCE

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Software Developer — *June 2023 – July 2023*

- Managed and optimized geospatial data layers in GeoServer for a production WebGIS application; performed data classification and reporting with QGIS
- Redefined the rendering pipeline and resolved performance bugs to reduce load time, while shipping new data-classification features

Indian Institute of Remote Sensing (ISRO) — Dehradun, India
Research Intern — *Feb. 2025 – May 2025*

- Analyzed and benchmarked multiple model architectures on RGB drone imagery, evaluating results via IoU and Dice score to inform which approach to recommend
- Built the full annotation-to-evaluation data pipeline (LabelMe → pixel masks → training/eval), presenting comparative findings across variants

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Research Summer Engineer — *May 2024 – July 2024*

- Devised a Physics-Informed Gaussian Process Regression model to forecast flood events, embedding domain constraints for improved reliability over a standard baseline
- Executed uncertainty quantification via Monte Carlo simulation, translating technical output into confidence-bounded risk predictions for decision-makers

PROJECTS

SolarWatch GB: Solar Generation Analysis & Reporting | *Python, MySQL, Docker*

- Built a MySQL pipeline joining PV_Live regional solar generation with Open-Meteo irradiance (323 UK regions × 60 days, 465K hourly rows), with an idempotent loader and a region-to-coordinate reconciliation that mapped 96.7% of regions.
- Flagged under-performing regions against weather-adjusted expectations (4.9% of region-days) and checked summed regional output against PV_Live's national total (98.9% coverage), since no ground truth exists.
- Published an interactive HTML dashboard with regional drill-downs, capacity factor, and trend panels; 23 pytest tests on the pure logic.

Telecom Churn Analysis & A/B Test Design | *Python, SQL, scikit-learn, SHAP*

- Uncovered a Simpson's paradox in synthetic customer data: a support add-on looked 6.5pp worse on raw churn but lowered churn in every risk quartile once stratified, because high-risk customers self-select into it.
- Segmented customers into 4 behavioral groups (17% to 60% churn) and used SHAP to show experience signals (dropped sessions, complaints) outrank contract type and tenure; checked on real IBM Telco data, where the paradox did not replicate.

ChainSignal: Crypto Market & On-Chain Anomaly Detection | *Python, SQL, ClickHouse, PySpark*

- Cleaned 1,000 coins × 365 days of market data in SQL views (excluded 83 feed-error coins, removed 921 one-day volume collapses) and built an interactive dashboard to explore flagged anomalies per coin.
- Found Celer Network trading $1.5B on an $18M market cap with flat price, confirmed in a second aggregator, as a wash-trading candidate; documented limitations such as survivorship bias and missed price-neutral exploits.

DuckDB vs. PostgreSQL: Analytical Query Performance Study | *DuckDB, PostgreSQL, SQL*

- Benchmarked DuckDB's columnar, vectorized OLAP engine against PostgreSQL's row-store on identical analytical queries over a multi-million-row dataset
- Used EXPLAIN / EXPLAIN ANALYZE to compare query execution plans, attributing DuckDB's speedup on aggregation-heavy queries to columnar storage and vectorized execution

Book Recommender — item-based collaborative filtering (cosine kNN on Book-Crossing) with a same-author content row, built on a course codebase *(Python, scikit-learn, Streamlit)*

Interactive Diamonds Data Analysis & Visualization — Power BI dashboard with drill-down visualizations analyzing pricing and quality attributes across a large retail dataset *(Power BI, Power Query)*

ACHIEVEMENTS

- Second Runner-Up, UCLA Glitch × Google DeepMind Hackathon — built BrandSync, then developed it fully after the Google team's positive feedback
- 1st Place, Inter-College Hackathon (Mental Health track) — built a Figma-prototyped app with an integrated chatbot
- Dean's List for Academic Excellence, Manipal University Jaipur (awarded twice)